# Vertical Jump Coach Agent — Progress Report

WatSPEED Agentic AI · Course Project · Sean Tang · due Nov 8, 2026
Code: github.com/seantanger/thp-jump-coach-agent (branch `main`)

For this project, I am building an AI coach for vertical jump training based on the information from John Evans / THP Strength YouTube video transcripts. The work I have done is with the help of Claude Code. The agents use the Claude API. The first thing I did was collect all of the transcripts from YouTube using yt-dlp, and preprocess the text with regex rules. I then tried three chunking techniques: fixed-size character splitting (LangChain's `RecursiveCharacterTextSplitter`) at two chunk sizes, and a third method that cuts the transcript into ~60-second windows at natural pauses in speech. After embedding the chunks using nomic embed in Ollama, I made a simple LangGraph loop where the Coach calls a search tool itself (up to 4 searches, rewriting its query if needed) and answers with the video and timestamp. I compared each of the three chunking methods with three search methods (keyword/BM25, vector, and a hybrid of the two) on 30 test questions. The best combination was the larger character chunks with vector search. I also built a second agent, a Planner, that writes multi-week training programs from reviewed templates, runs them through code checks, and waits for my approval before saving (this is still a WIP).

I went through a few versions of the Coach prompt, most recently to make answers more concise. For safeguards, a code check rejects any answer that cites a source the search didn't return, a third agent (the Verifier) checks each point against its source in a fresh context, and a code safety check sends messages with red-flag pain symptoms (like a pop in the knee) to a physio instead of coaching them.

Next Steps/Ideas:

- Plan quality: reading the first plan, sessions needed more detail. Month 1 accessories are now named exercises (the template lists options such as RDL, hip thrust and Nordic curls). Still to do: the same for the general-circuit days and months 2–6, a load for every lift, and warm-up and ramp-up sets instead of one weight for every set. I also need to still work on the re-editing plan process. For example, allow the user to keep reiterating the plan if they have suggestions.
- Connect the Planner and Coach. Right now they are two separate LangGraph graphs: a code router detects plan requests, but the chat doesn't pass them to the Planner yet. I want to maybe combine them into one parent graph with the Coach and Planner as subgraphs.

- Give the Coach access to the current plan and profile, so its answers fit where the athlete is in the program.
- Memory: right now the profile and each plan version are saved as JSON files. Would something like Letta be better?
- Training log: record sessions as done, changed or missed, plus pain and jump results, and use them to decide when the athlete moves to the next stage. (Not sure if this should be like a UI or the user should enter a chat to tell the system that they have finished a workout)
- Workout variety: the Planner can now swap exercises (at most 2 per week, each cited and equipment-checked); next is more options per slot and loosening sets and reps.
- Optional: MCP server.

## Demos

These call the Claude API (about 4¢ for the Coach question, about 25¢ for a plan) and read the transcript index. The outputs are saved in this notebook.

In [1]:
```
import os, sys
from pathlib import Path

ROOT = (Path.cwd().parent if Path.cwd().name == "notebooks" else Path.cwd())
os.chdir(ROOT); sys.path.insert(0, str(ROOT))
os.environ.setdefault("RETRIEVAL_MODE", "dense")  # dense (vector) search ne
```

Out[1]:
```
'dense'
```

**The Coach's LangGraph setup** ( `src/coach/graph.py` ). Two nodes: `coach` (the LLM with tools bound) and `tools` (runs the searches, capped at 4). The graph loops coach → tools → coach until the LLM stops calling tools. `answer_question` (used below) builds this graph, runs it, checks the citations, and sends the answer to the Verifier.

**1. Ask the Coach a question.** The Coach searches the library, answers with citations, then the Verifier checks it. (The two agents are linked through code, not the LangGraph)

In [2]:
```
from src.coach.qa import answer_question

r = answer_question("Should I static stretch before jumping?")
print(r["compact"])
print("\nsearches:", r["queries"])
print("citation check passed:", r["check"]["passed"], "| Verifier:", r["chec
```

```
Generally no. Skip static stretching right before jumping and use dynamic fl
exibility instead. If you need it for mobility, do short holds and follow th
em with something dynamic or ballistic, or do them after the session.

- Static stretching before jumping can temporarily lower force and power out
put, so keep the pre-jump warm-up dynamic (RDLs, side squats, leg swings, lo
ng lunges). [1][2]
- The power drop is dose-dependent. Long, intense stretching hurts most, and
30–60 seconds followed by something ballistic largely cancels the negative e
ffect. [3][4]
- If tight ankles or hips limit your squat or jump positions, do static stre
tches after your warm-up, or in the evening. [5][4]

**Sources**
1. [John Evans — A Practical 2 Day Per Week Jump System Guaranteed To Get Yo
u Dunking @ 01:42](https://www.youtube.com/watch?v=1sQ-FOxDP5M&t=102s)
2. [THP Strength — World Record Vertical Jumper's Exact Warm-Up Routine (50.
5 Inches) @ 00:51](https://www.youtube.com/watch?v=O4piVLyfvSA&t=51s)
3. [John Evans — Does Stretching ACTUALLY Make You A Better Athlete? What Do
es Science Say? @ 00:00](https://www.youtube.com/watch?v=ifrQ8lcxoNs&t=0s)
4. [THP Strength — This Test Will Show You If You Can Reach A 40 Inch Vertic
al @ 07:44](https://www.youtube.com/watch?v=8l0Omkm28hE&t=464s)
5. [John Evans — A Practical 2 Day Per Week Jump System Guaranteed To Get Yo
u Dunking @ 05:42](https://www.youtube.com/watch?v=1sQ-FOxDP5M&t=342s)

searches: ['static stretching in the warm-up before jumping', 'static stretc
hing effect on jump performance power output']
citation check passed: True | Verifier: pass
```

A pain question: the answer sticks to what the library says about tendon pain, says what it doesn't cover, and ends with the physio line. Note that regular AI models typically recommend ice and anti inflammatories which are NOT recommended by THP.

In [3]:
```
r = answer_question("What should I do if I have knee pain?")
print(r["compact"])
print("\nsearches:", r["queries"])
print("citation check passed:", r["check"]["passed"], "| Verifier:", r["chec
```

```
If your knee pain is activity-related tendon ache under the kneecap, take 2
4–48 hours off, then track your morning pain. If it stays above 3/10, start
daily isometrics.

- For an early or reactive flare-up, taking 24 to 48 hours off often settles
the tendon. You can then return to normal activity without new exercises.
[1]
- Test pain each morning with a deep bodyweight squat, scored 1–10. Pain pea
ks 24–48 hours after a jump session. If the score is above 3/10, start isome
trics. [2][3]
- Isometrics: hold a quad exercise such as a leg extension for 45 seconds at
about 70% effort. Do 3–5 sets, about three times a day, spaced roughly 6 hou
rs apart. [4][5]

*See a physio if pain is sharp/stabbing or steadily worsening, occurs at res
t or wakes you at night, lasts over two weeks without improving despite cutt
ing load, or follows a fall/impact; or if you have swelling, bruising, defor
mity, a pop, giving way, locking, catching, numbness, tingling, radiating we
akness or limping. Back pain with bladder/bowel changes or groin numbness ne
eds emergency care.*

**Sources**
1. [THP Strength – How to Fix Patellar Tendon Pain Based on Your Tendinopath
y Phase @ 01:24](https://www.youtube.com/watch?v=iLI8hXsB6hs&t=84s)
2. [THP Strength – How to Fix Patellar Tendon Pain Without Quitting Your Spo
rt @ 03:07](https://www.youtube.com/watch?v=ZN8wh4FFZuA&t=187s)
3. [THP Strength – The Exact Protocol to Fix Knee Pain and Get Back to Jumpi
ng @ 00:00](https://www.youtube.com/watch?v=VMSF8C25mxY&t=0s)
4. [John Evans – How To Get Rid Of Knee Pain (Full Blueprint) @ 05:28](http
s://www.youtube.com/watch?v=9X_j-oF2ZDg&t=328s)
5. [THP Strength – Knee Pain Killing Your Vertical? Start With Isometrics @
02:17](https://www.youtube.com/watch?v=TMznIxcRJGg&t=137s)

searches: ["patellar tendon pain tendinopathy jumper's knee what to do", 'kn
ee pain training load modification rehab isometrics']
citation check passed: True | Verifier: pass
```

**2. Safety check (code, no LLM).** Red-flag symptoms get a physio referral before any agent runs; plan requests are flagged for the Planner (not yet wired into the chat).

In [4]:
```python
from src.red_flags import referral_kind
from src.router import route
```

In [5]:
```python
for m in ["Should I static stretch before jumping?", "I felt a pop in my kne
          "Make me a plan for the next 12 weeks"]:
    print(f"{m:42} referral={referral_kind(m)}  route={route(m)}")
```

```
Should I static stretch before jumping?    referral=None  route=coach
I felt a pop in my knee this week          referral=physio  route=coach
Make me a plan for the next 12 weeks       referral=None  route=program
```

**3. First plan from the Planner.** Needs a profile first (`uv run python -m scripts.plan profile`). The Planner drafts a plan, the code checks and Verifier run, then the graph pauses for my approval.

**The Planner's LangGraph setup** ( `build_plan_graph` in `src/planner/graph.py` ). `context` loads the profile and current plan, `planner` ⇄ `tools` is the LLM tool loop, `review` runs the 8 code checks and the Verifier (failures go back to the planner, at most 2 retries), `approve` pauses the graph until I approve, reject or revise, and `save` writes a new plan version.

**The athlete profile (me).** This is what the Planner reads with its `get_profile` tool. It's stored as `profile.json` and made with `uv run python -m scripts.plan profile` .

In [6]:
```
from src.planner.store import load_profile

print(load_profile().model_dump_json(indent=2))
```

```
{
  "training_age_years": 4.0,
  "lifts": {
    "back_squat_lb": 225.0,
    "half_squat_lb": 245.0,
    "power_clean_lb": 175.0
  },
  "standing_vertical_in": null,
  "approach_vertical_in": 32.0,
  "standing_reach_in": 86.5,
  "height_in": 67.0,
  "days_per_week": 7,
  "available_days": [
    "mon",
    "tue",
    "wed",
    "thu",
    "fri",
    "sat",
    "sun"
  ],
  "session_minutes": 90,
  "equipment": [
    "barbell",
    "rack",
    "platform",
    "box",
    "med ball",
    "safety bar",
    "belt squat",
    "bench",
    "hamstring curl machine"
  ],
  "event": null,
  "goal": "Dunk off two feet. Approach vertical about 31–33 in, standing rea
ch about 7 ft 2–3 in (needs roughly 10 ft 6 in touch). Back squat is a deep
squat; half squat estimated.",
  "knee_pain": {
    "status": "ache",
    "notes": "Knee pain in the past, now manageable."
  },
  "injury_notes": "Two-foot jumper. No sport or other jump load during the w
eek."
}
```


In [7]:
```
from src.planner import run
from src.planner.render import render_proposal

plan = run.start("new")
print(render_proposal(plan))
```

```
What changes:
  New plan outline
  Month 1: volume
  Month 2: intensification (power and 6–8 rep strength)
  Month 3: absolute strength
  (+5 more)
Why: New 6-month long-conjugate outline of six mesocycles, running 9 Nov 202
6 to 25 Apr 2027. Weeks 1–2 of Month 1 are written day by day (9–22 Nov). Ea
ch week has lifts Mon/Wed/Fri, general days Tue/Thu and a jump session on Sa
t.
⚠ weeks[2026-11-09]: partly supported — Main lifts, jump 35+5 min and isomet
rics are supported. Accessory 3x10 and circuit details are not stated in the
cited chunks.
```

In [8]:
```python
plan = run.resume(plan.thread_id, "approve")  # or "reject", or "revise" wit
print(render_proposal(plan))
```

```
Saved as version 1.
```

**The saved plan in detail.** The 6-month outline and every week written so far (the Planner writes 2 weeks at a time and rolls forward), then the videos the plan draws on. Every session cites the transcript chunks it came from; those are hidden here and shown in the Developer view. (Note that this is still a WIP)

In [9]:
```python
from src.kb.evidence import chunk_meta
from src.planner.render import render_outline, render_week
from src.planner.store import load_program
from src.planner.templates import TEMPLATE_METHOD

program = load_program()
print(f"Plan version {program.version}: {program.change_summary}\n")
print(render_outline(program) + "\n")
for week in program.weeks:
    print(render_week(program, week.start) + "\n")

cites = {c for m in program.mesocycles for c in m.cites} | {
    c for w in program.weeks for d in w.days for s in d.sessions for c in s.
videos = {}  # one line per video the plan draws on
for meta in chunk_meta(cites, method=TEMPLATE_METHOD).values():
    videos.setdefault(meta["title"], meta["url_t"])
print("Videos used:")
for title, url in sorted(videos.items()):
    print(f"  {title}  {url}")
```

Plan version 1: New 6-month long-conjugate outline of six mesocycles, runnin g 9 Nov 2026 to 25 Apr 2027. Weeks 1–2 of Month 1 are written day by day (9– 22 Nov). Each week has lifts Mon/Wed/Fri, general days Tue/Thu and a jump se ssion on Sat.

Month 1 · Mon Nov 9 – Sun Dec 6 · volume · Build work capacity with general work that looks little like jumping; unload in week 4.
Month 2 · Mon Dec 7 – Sun Jan 3 · intensification (power and 6–8 rep strengt h) · Intensify the month-1 volume with clean pulls and power cleans; heavier RDL and calf raise accessories prime month 3.
Month 3 · Mon Jan 4 – Sun Jan 31 · absolute strength · Heavy squats and heav y clean pulls; eccentric accessories prime month 4.
Month 4 · Mon Feb 1 – Sun Feb 28 · strength-speed (fast bars, eccentric over load) · Power clean to maxes; accentuated eccentric squats and eccentric spl it squats; general days get more general.
Month 5 · Mon Mar 1 – Sun Mar 28 · speed-strength (bar speed) · Move bars fa st (lighter cleans, power snatch, hang cleans); blend a heavy squat primer w ith high-velocity squats; start shock loading at low plyo volume.
Month 6 · Mon Mar 29 – Sun Apr 25 · specificity · Specificity at its highes t, with the most plyo intensity and variety of the year; accessories become general and slow.

Month 1 of 6 · volume · Week 1 of 4

Mon  power clean 5×4 · back squat 5×6 @ 70% · RDL 3×10
      power clean: Power exercise first. Do 3x45 s isometrics before the ses sion.
      back squat: Percent of your 225 lb back squat (about 158 lb).
      RDL: Hip hinge; general accessory.
Tue  general circuit (varied movements)
      general circuit (varied movements): Bodyweight or bodybuilding circui t, upper body, core. Nothing that looks like jumping.
Wed  power clean 5×4 · back squat 4×12 @ 65% · hip thrust 3×10
      power clean: Power exercise first. Do 3x45 s isometrics before the ses sion.
      back squat: Percent of your 225 lb back squat (about 146 lb).
      hip thrust: Hip-extension accessory; low risk.
Thu  general circuit (varied movements)
      general circuit (varied movements): Keep it different from Tuesday's c ircuit. Easy and general.
Fri  power clean 5×4 · back squat 4×8 @ 68% · lying hamstring curl 3×10
      power clean: Power exercise first. Do 3x45 s isometrics before the ses sion.
      back squat: Percent of your 225 lb back squat (about 153 lb).
      lying hamstring curl: Easy to add at the end; barely taxing.
Sat  jump session warm-up jumps @ 35 min · jump session max-effort jumps @ 5 min
      jump session warm-up jumps: Low-effort jumping inside the 40 min windo w. Do 3x45 s isometrics before the session.
      jump session max-effort jumps: Max-effort jumps. Stop early if knee pa in goes above 2–3/10 while jumping.

ⓘ sources

Month 1 of 6 · volume · Week 2 of 4

Mon  power clean 5×4 · back squat 5×6 @ 70% · Nordic hamstring curl 3×8
      power clean: Power exercise first. Do 3x45 s isometrics before the ses
sion.
      back squat: Percent of your 225 lb back squat (about 158 lb). Try to a
dd a little load or a rep versus last week.
      Nordic hamstring curl: Knee-flexion option; eccentric emphasis.
Tue  general circuit (varied movements)
      general circuit (varied movements): Bodyweight or bodybuilding circui
t, upper body, core. Nothing that looks like jumping.
Wed  power clean 5×4 · back squat 4×10 @ 67% · RDL 3×10
      power clean: Power exercise first. Do 3x45 s isometrics before the ses
sion.
      back squat: Percent of your 225 lb back squat (about 151 lb).
      RDL: Hip hinge; general accessory.
Thu  general circuit (varied movements)
      general circuit (varied movements): Keep it different from Tuesday's c
ircuit. Easy and general.
Fri  power clean 5×4 · back squat 4×8 @ 70% · hip thrust 3×10
      power clean: Power exercise first. Do 3x45 s isometrics before the ses
sion.
      back squat: Percent of your 225 lb back squat (about 158 lb).
      hip thrust: Hip-extension accessory; low risk.
Sat  jump session warm-up jumps @ 35 min · jump session max-effort jumps @ 5
min
      jump session warm-up jumps: Low-effort jumping inside the 40 min windo
w. Do 3x45 s isometrics before the session.
      jump session max-effort jumps: Max-effort jumps. Stop early if knee pa
in goes above 2–3/10 while jumping.

ⓘ sources

Videos used:
  3 Best Exercises To Jump Higher In 2025  https://www.youtube.com/watch?v=O
AsUemGRrpw&t=891s
  Everything Athletes Ask Us About Jumping Higher  https://www.youtube.com/w
atch?v=3vmd30VvzRA&t=855s
  How To Get Rid Of Knee Pain (Full Blueprint)  https://www.youtube.com/watc
h?v=9X_j-oF2ZDg&t=288s
  How To Train Your Hamstrings To Jump Higher  https://www.youtube.com/watc
h?v=I1baSIlV_pQ&t=408s
  How to Get a 40" Vertical (Full Blueprint)  https://www.youtube.com/watch?
v=Icml891MLHA&t=141s
  My Full Weekly Vertical Jump Training Routine Explained  https://www.youtu
be.com/watch?v=SXjO279Oslw&t=137s
  You'll Never Get Hurt Again If You Understand These Principles  https://ww
w.youtube.com/watch?v=72YX7nkYmYE&t=142s